Shravani Anil Patil
D20A - 40

Experiment: 07

Aim: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

MetaMask is a **cryptocurrency wallet and gateway** to blockchain applications. It allows users to store, send, and receive Ether or tokens, and connect to decentralized applications (DApps) directly from a web browser. MetaMask acts as a **bridge** between a user's browser and the Ethereum blockchain (or other compatible networks).

Features**:**

- Easy management of multiple blockchain accounts
- Secure private key storage
- Integration with test networks like **Sepolia**, **Linea**, or **RSK**
- Interaction with DApps and smart contracts via **Remix IDE**

Testnet:

A Testnet is a blockchain network used for testing and development without using real cryptocurrency. It mimics the main Ethereum network but uses test Ether (fake ETH). Common testnets include Sepolia Testnet, Linea Testnet, and RSK Testnet.

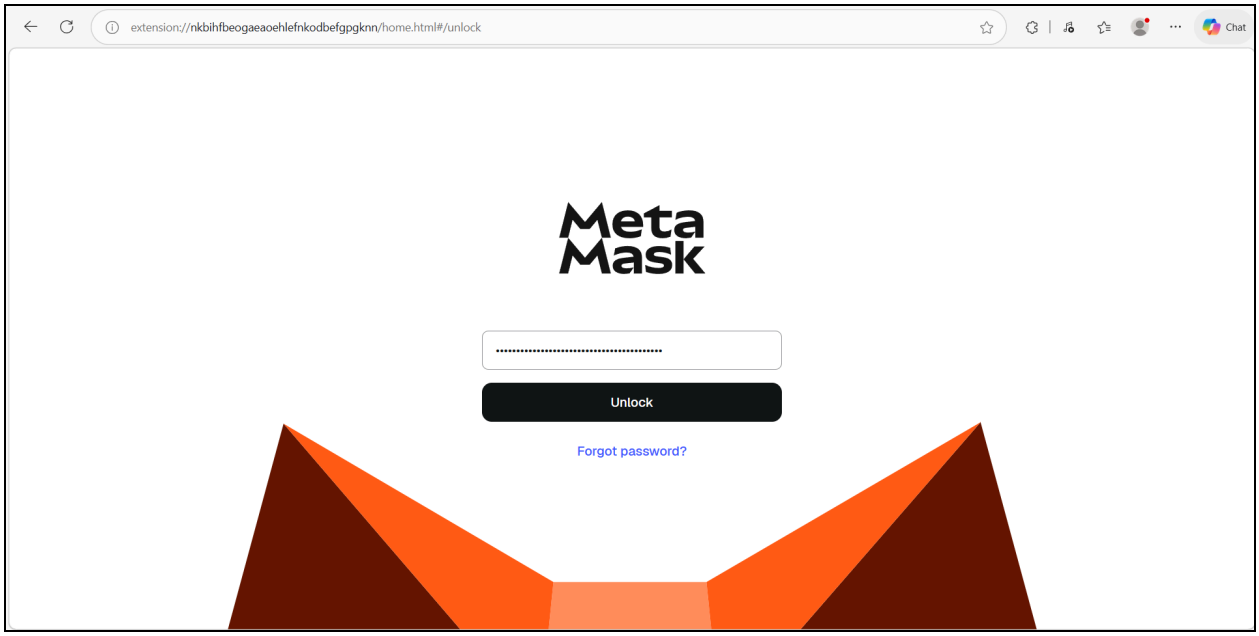
Advantages:

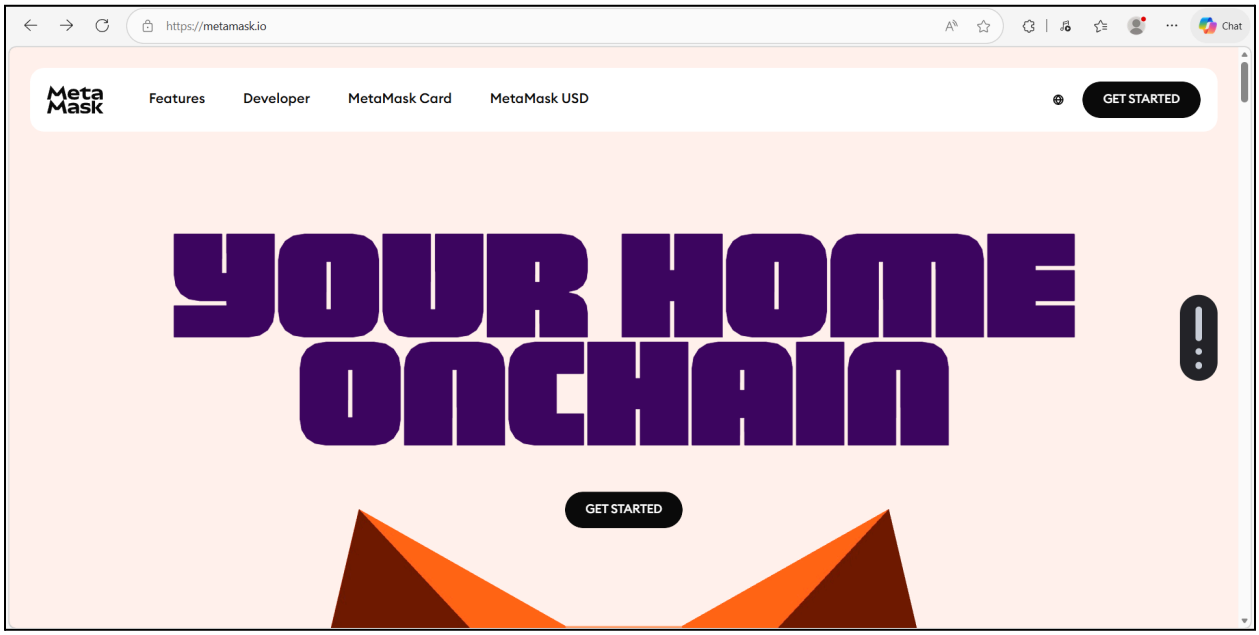
- No real money loss during testing
- Safe environment for developers
- Helps test smart contracts and transactions before mainnet deployment

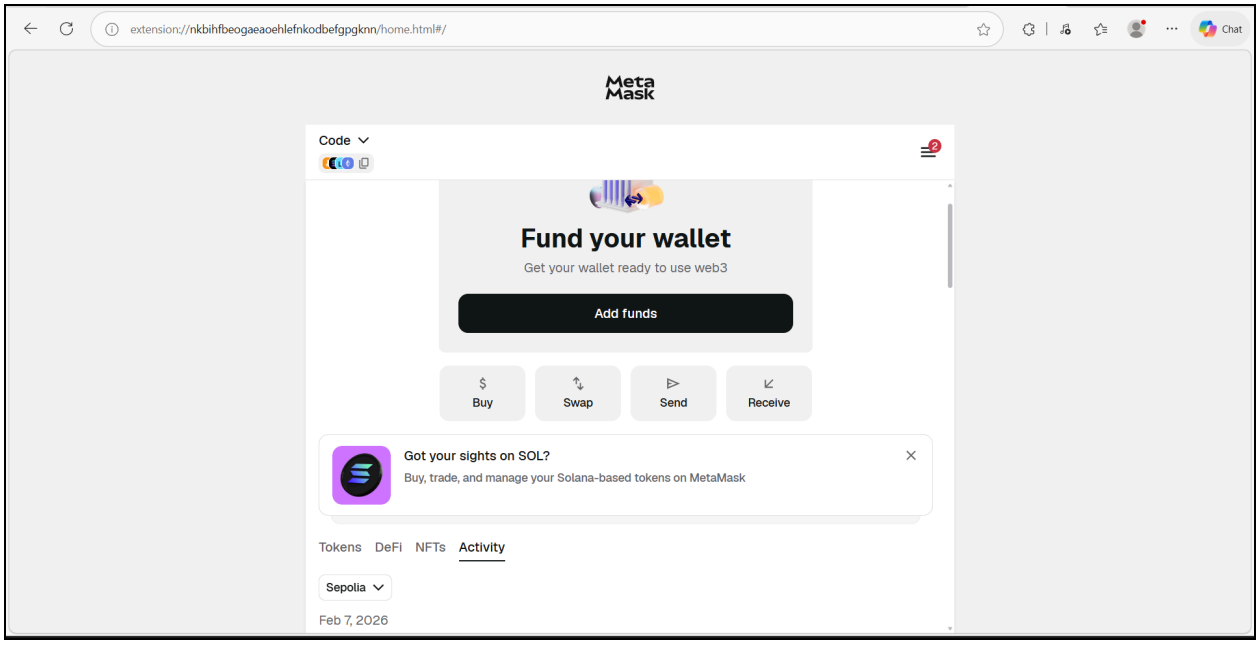
Steps to Connect MetaMask with Remix IDE for Performing Transactions:

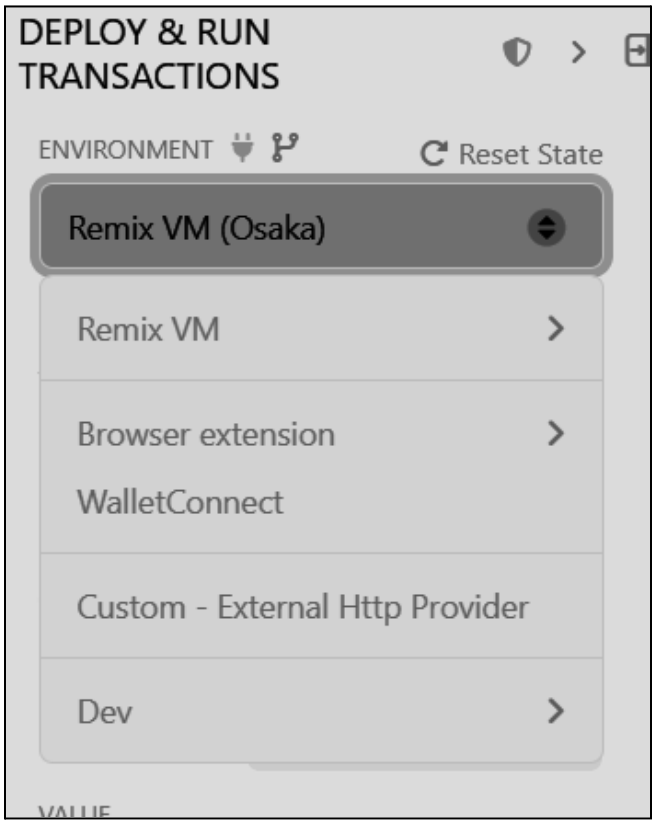
1. Set Up MetaMask:
    - Install the MetaMask extension on your browser.
    - Create or import a wallet account using a recovery phrase.
2. Add a Test Network:
    - Open MetaMask → Settings → Networks → Add Network.
    - Enter details for the desired Testnet (e.g., Sepolia or RSK).
    - Example RPC URL: https://sepolia.infura.io/v3/<your_project_id>
3. Fund the Wallet:
    - Use a faucet to receive free test Ether (e.g., 0.5 ETH/day).
    - Check the balance in MetaMask.

4. Connect MetaMask with Remix IDE:
    - Open Remix IDE.
    - In the Deploy & Run Transactions tab, choose Injected Web3 as the environment.
    - Approve connection in MetaMask.


extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html#/unlock
Chat
Meta Mask
Unlock
Forgot password?

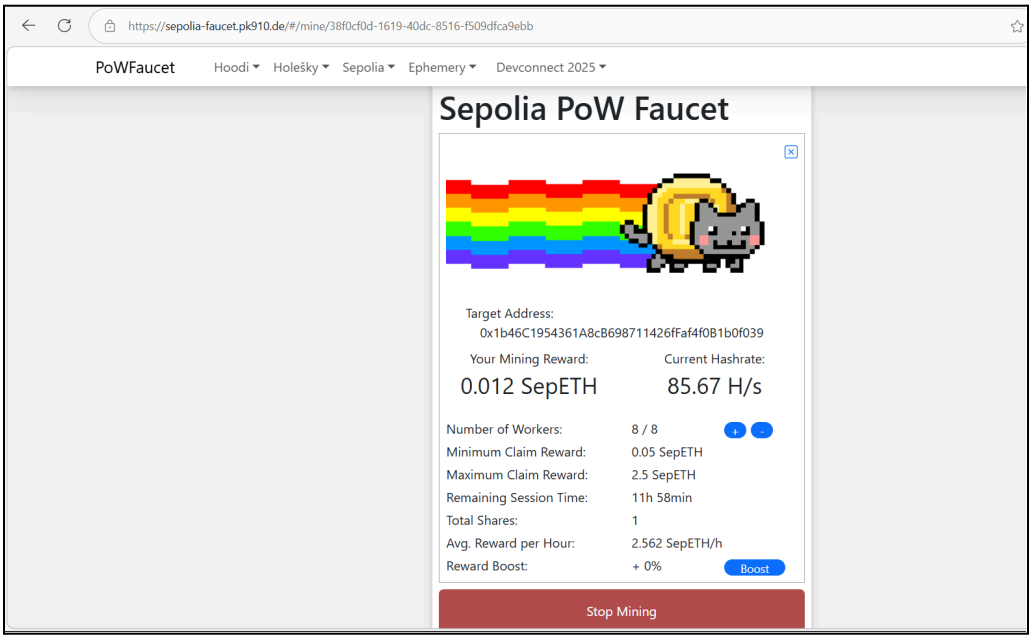

https://metamask.io
Chat
Meta Mask
Features
Developer
MetaMask Card
MetaMask USD
GET STARTED
YOUR HOME ONCHAIN
GET STARTED


extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html#/
Chat
Meta Mask
Code
Fund your wallet
Get your wallet ready to use web3
Add funds
Buy
Swap
Send
Receive
Got your sights on SOL?
Buy, trade, and manage your Solana-based tokens on MetaMask
Tokens
DeFi
NFTs
Activity
Sepolia
Feb 7, 2026

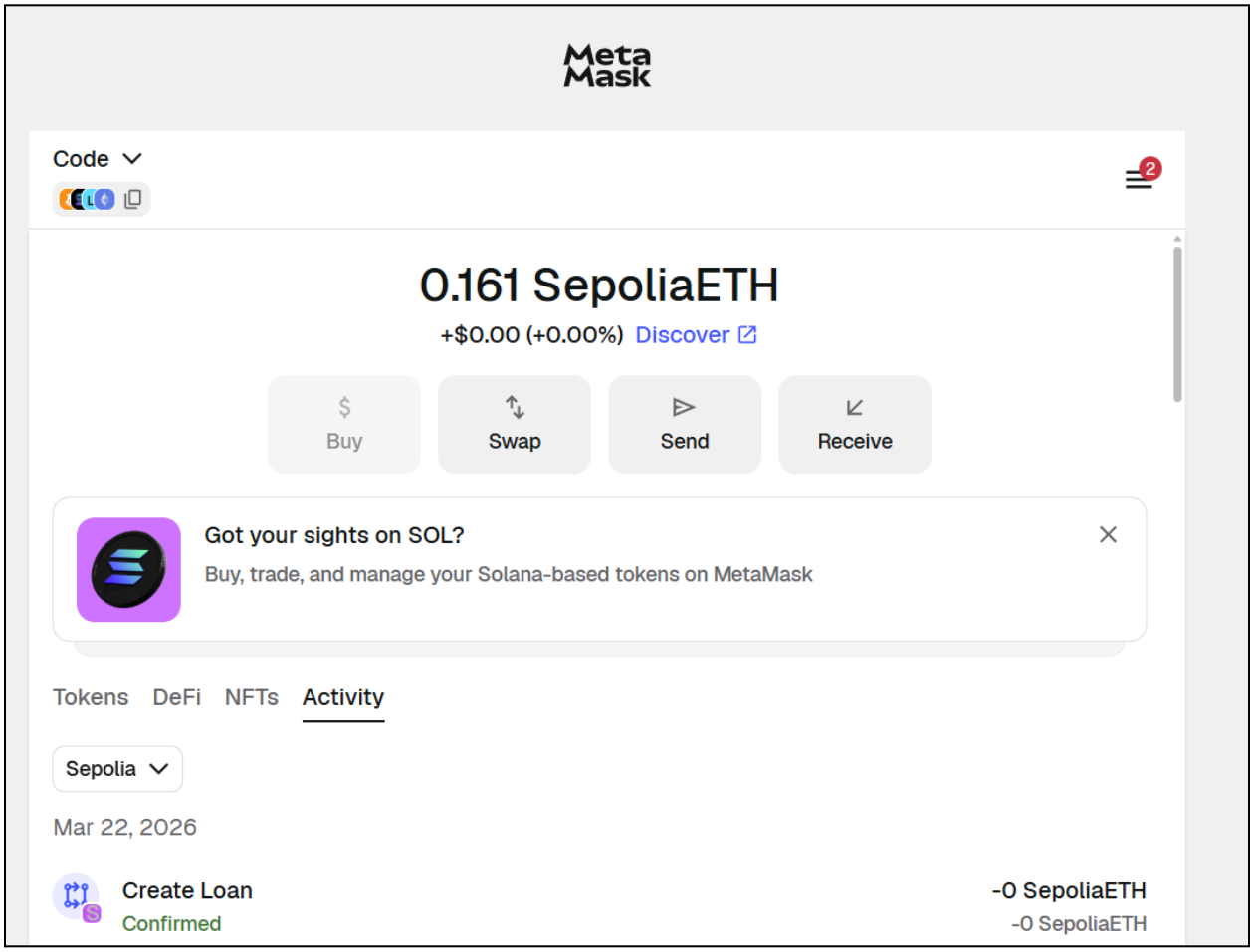

DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Reset State
Remix VM (Osaka)
Remix VM
Browser extension
WalletConnect
Custom - External Http Provider
Dev

Sepolia test network connected to RemixIDE using metamask

Sepolia Faucet Mining



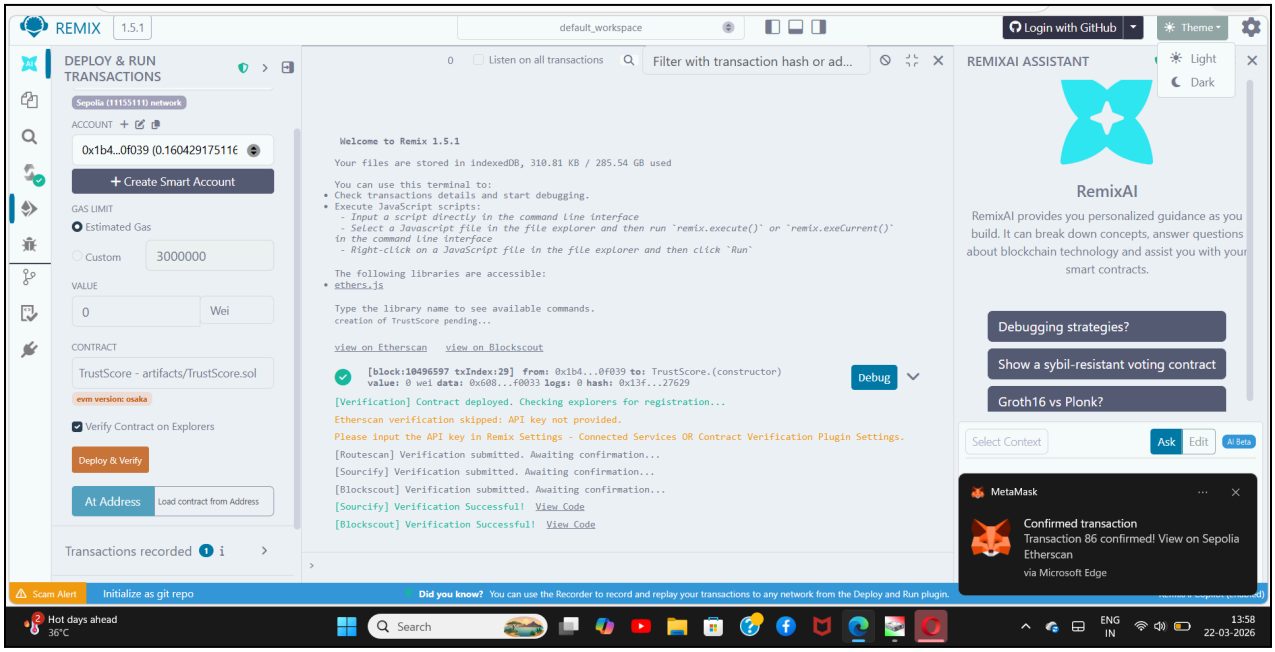
Account balance



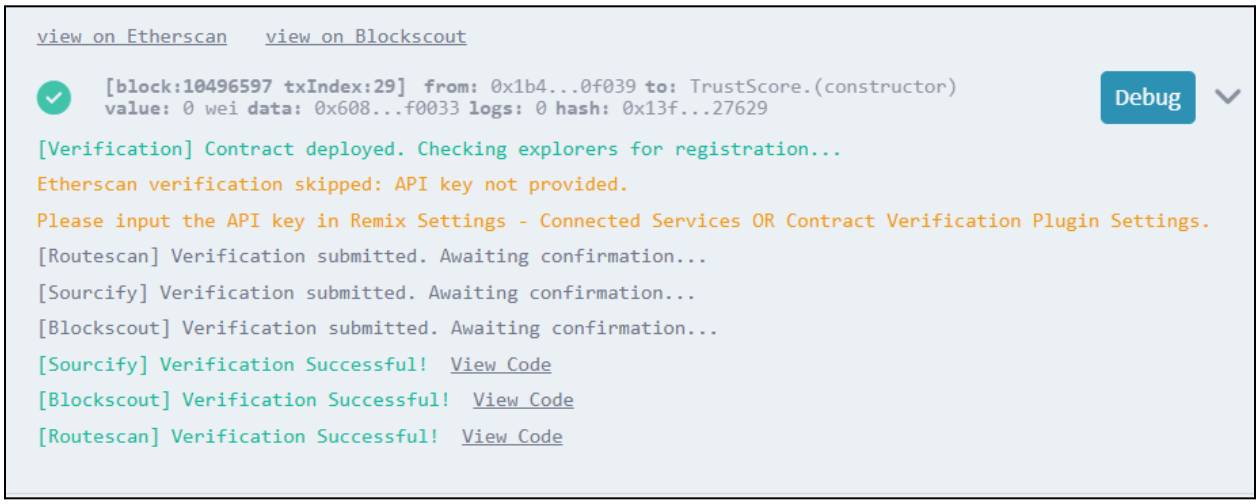
TrustScore.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract TrustScore {
    // Store trust scores for each user
    mapping(address => uint256) public trustScores;
    address public admin;
    address public loanManagerAddress;
    uint256 public initialScore = 50; // Initial score for new users

    constructor() {
        admin = msg.sender;
    }

    // Set LoanManager address (admin only)
    function setLoanManagerAddress(address _loanManagerAddress) public {
        require(msg.sender == admin, "Only admin");
```

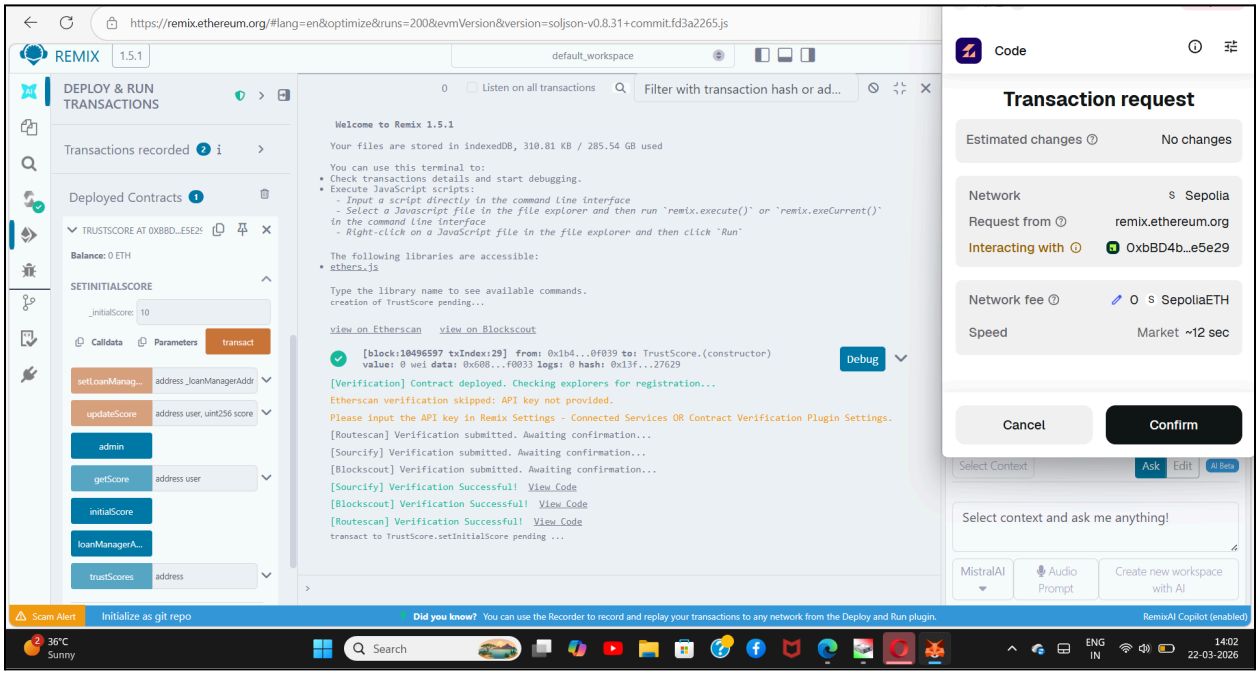
```
        loanManagerAddress = _loanManagerAddress;
    }

    // Admin or LoanManager can update scores
    function updateScore(address user, uint256 score) public {
        require(msg.sender == admin || msg.sender == loanManagerAddress, "Only admin or
LoanManager");
        require(score <= 100, "Score must be 0-100");
        trustScores[user] = score;
    }

    // Set initial score for new users (admin only)
    function setInitialScore(uint256 _initialScore) public {
        require(msg.sender == admin, "Only admin");
        require(_initialScore <= 100, "Score must be 0-100");
        initialScore = _initialScore;
    }

    // Anyone can check a score - returns initialScore if not set
    function getScore(address user) public view returns (uint256) {
        uint256 score = trustScores[user];
        // If score is 0, return initial score (user hasn't been explicitly set)
        return score > 0 ? score : initialScore;
    }
}
```



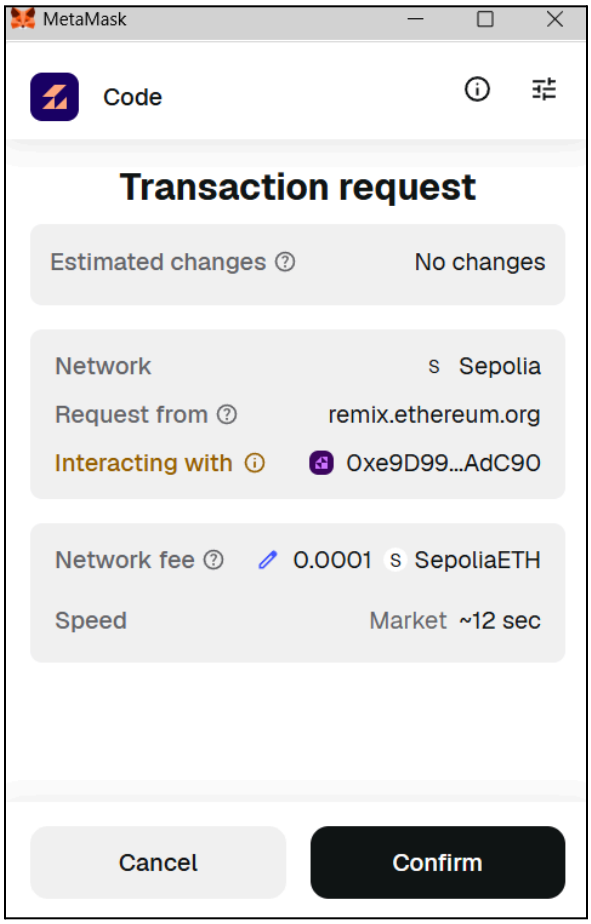
Contract Deployed:

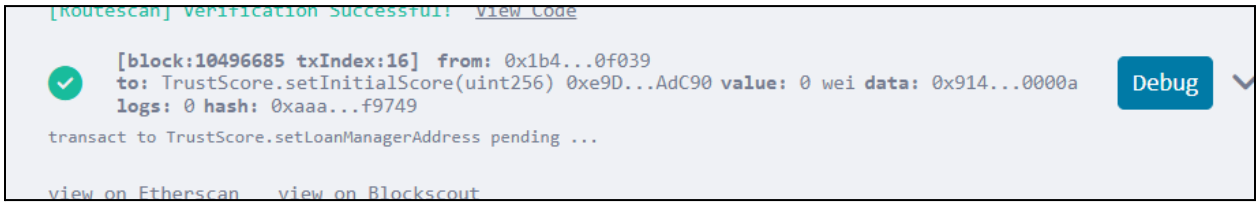


Transaction request

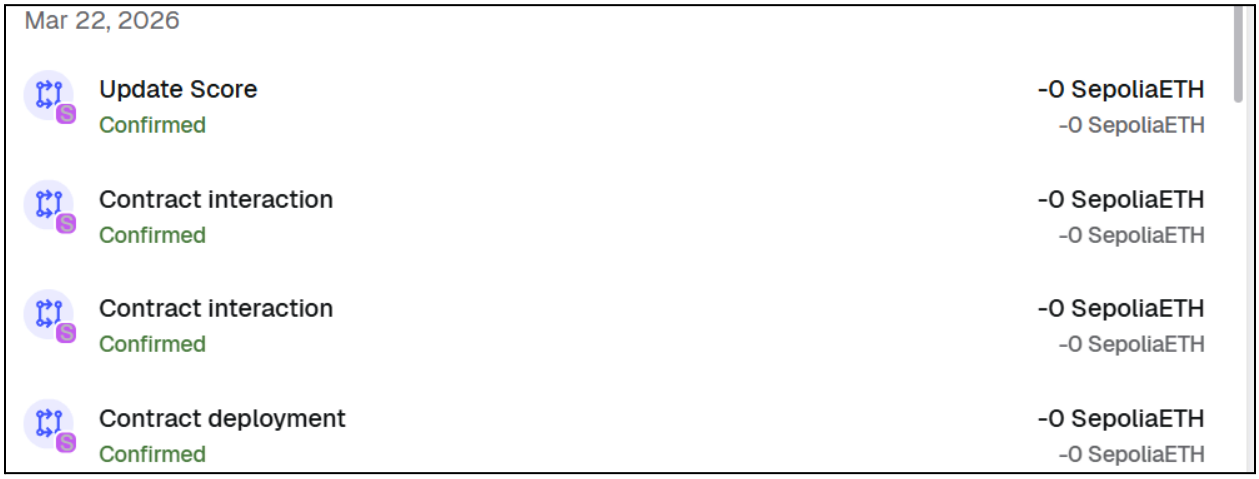


Confirmation on Metamask

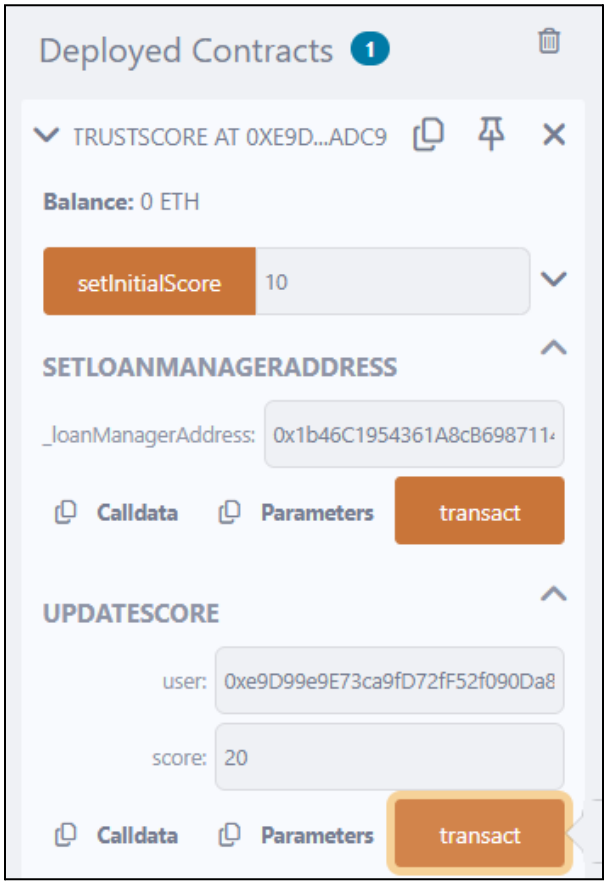


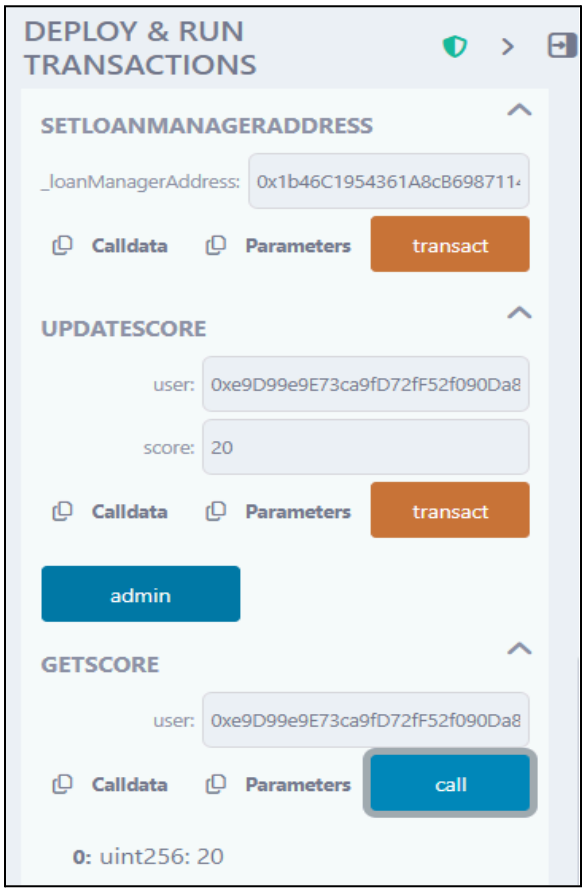


Contract details on Metmask:



Functions in contract executed :





Conclusion:

In this experiment, we successfully connected **Remix IDE** to the **Sepolia Testnet** using **MetaMask** as the wallet provider. After configuring the correct network and linking the wallet through the Injected Provider, Remix was able to interact directly with the blockchain.

Smart contracts were then compiled, deployed, and executed on the Sepolia network. The test ETH was used to pay gas fees, and all transactions were recorded on the blockchain, demonstrating a real-time decentralized execution environment.

This experiment provided practical exposure to blockchain development workflows, including wallet integration, testnet usage, contract deployment, and transaction validation, helping us understand how decentralized applications are developed and tested before mainnet deployment.